

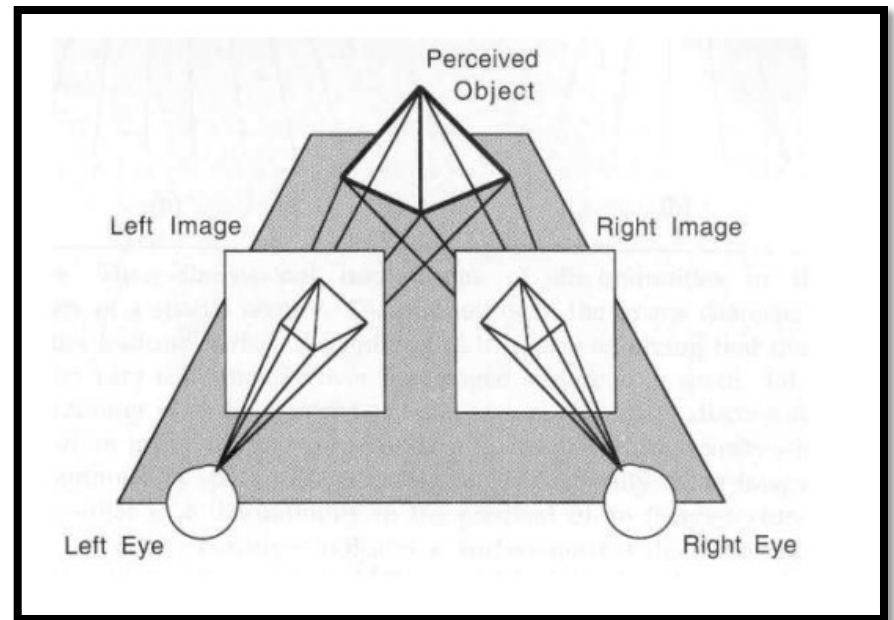
Stereo Reconstruction

Contents

- Stereo Vision
- Depth With Stereo
- Correspondence Search
- Dynamic Programming based Stereo Matching

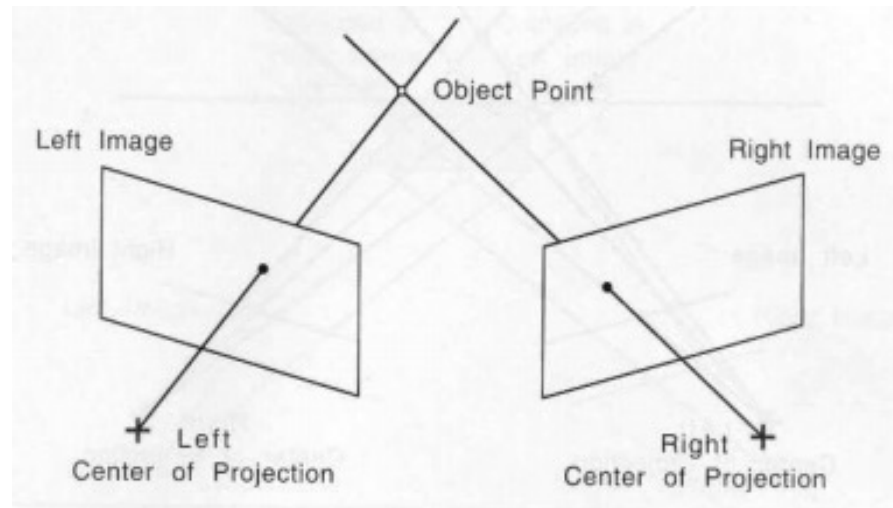
Stereo Vision – Goal

- The recovery of the 3D structure of a scene using two images acquired from a different viewpoint in space
- The images can be obtained using two cameras separated by a distance or one moving camera.



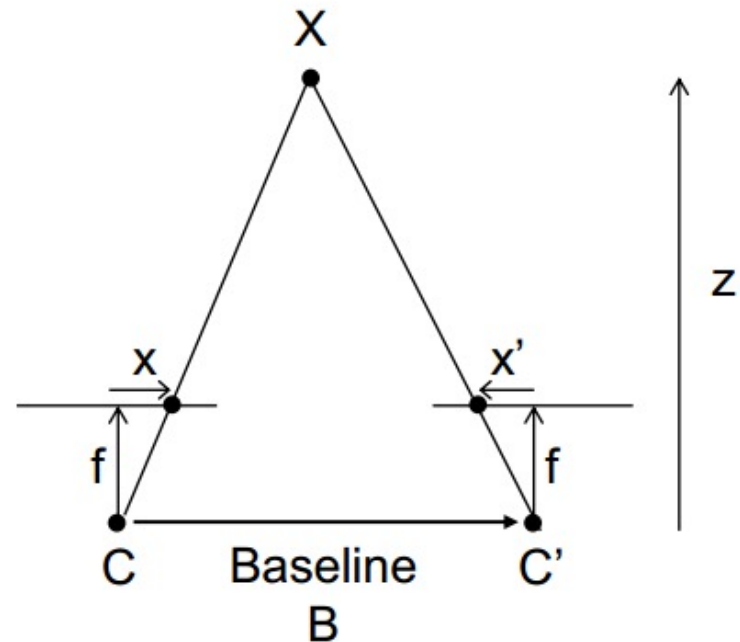
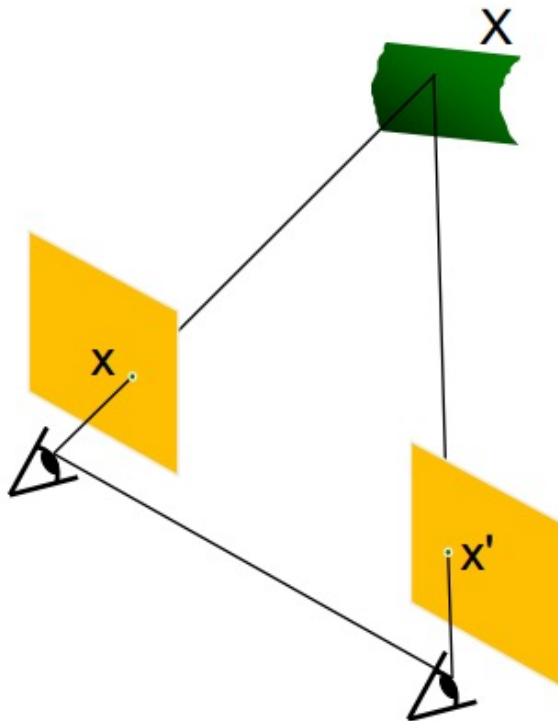
Triangulation – Principle underlying Stereo Vision

- The 3D location of point in space is restricted to straight line that passes through center of projection and the projection of the object point on image plane
- Stereo Vision determines the position of a point in space by finding the intersection of the two lines passing through center of projection and the projection of object point on image planes



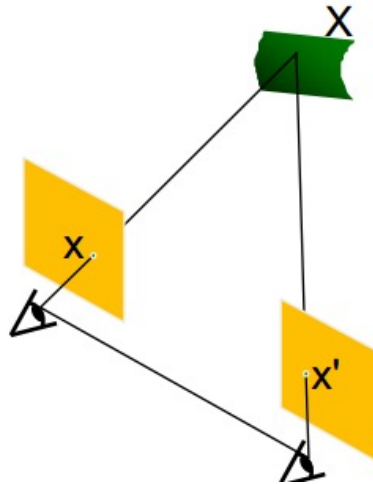
Stereo Vision - Notation

- recover depth by finding image coordinate x' in one image that corresponds to x in the other image



Stereo Vision – Sub problems

- Calibration: How do we recover the relation of the cameras – i.e the distance between camera centers, orientation of image planes w.r.t each other (If not already known?)
- Correspondence: How do we search for the matching point x' ?



Stereo Correspondence Problem



- We have two images taken from cameras with different intrinsic (focal length, resolution) and extrinsic (distance between cameras and orientation change) parameters
- How do we match a point in the first image to a point in the second?
How can we constrain our search?

Depth Map Estimation

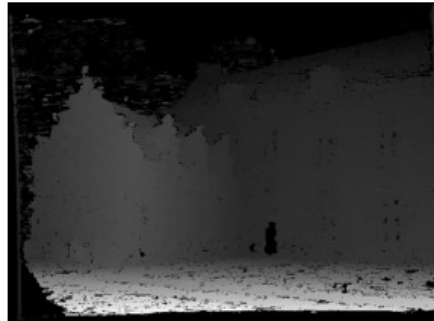
image 1



image 2



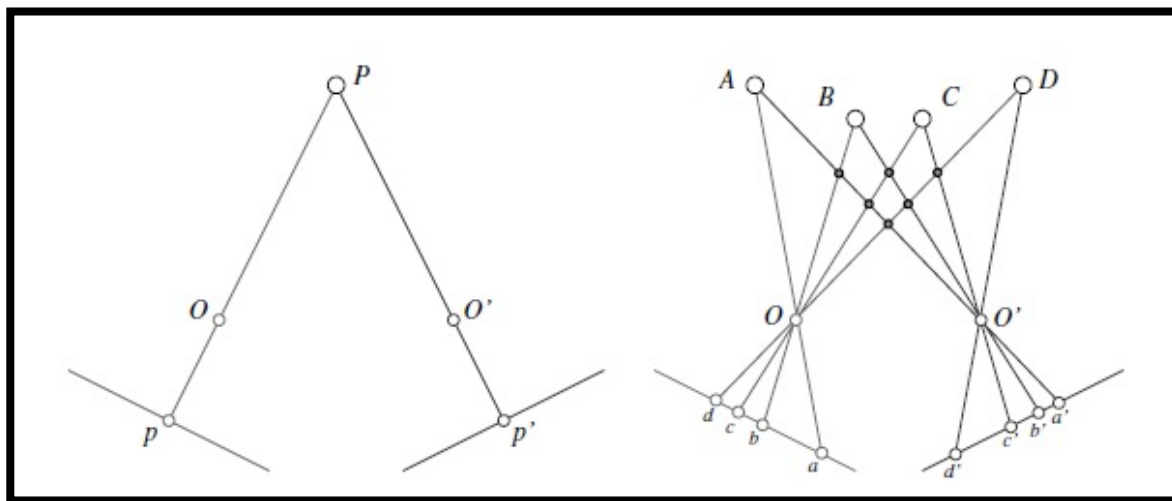
Dense depth map



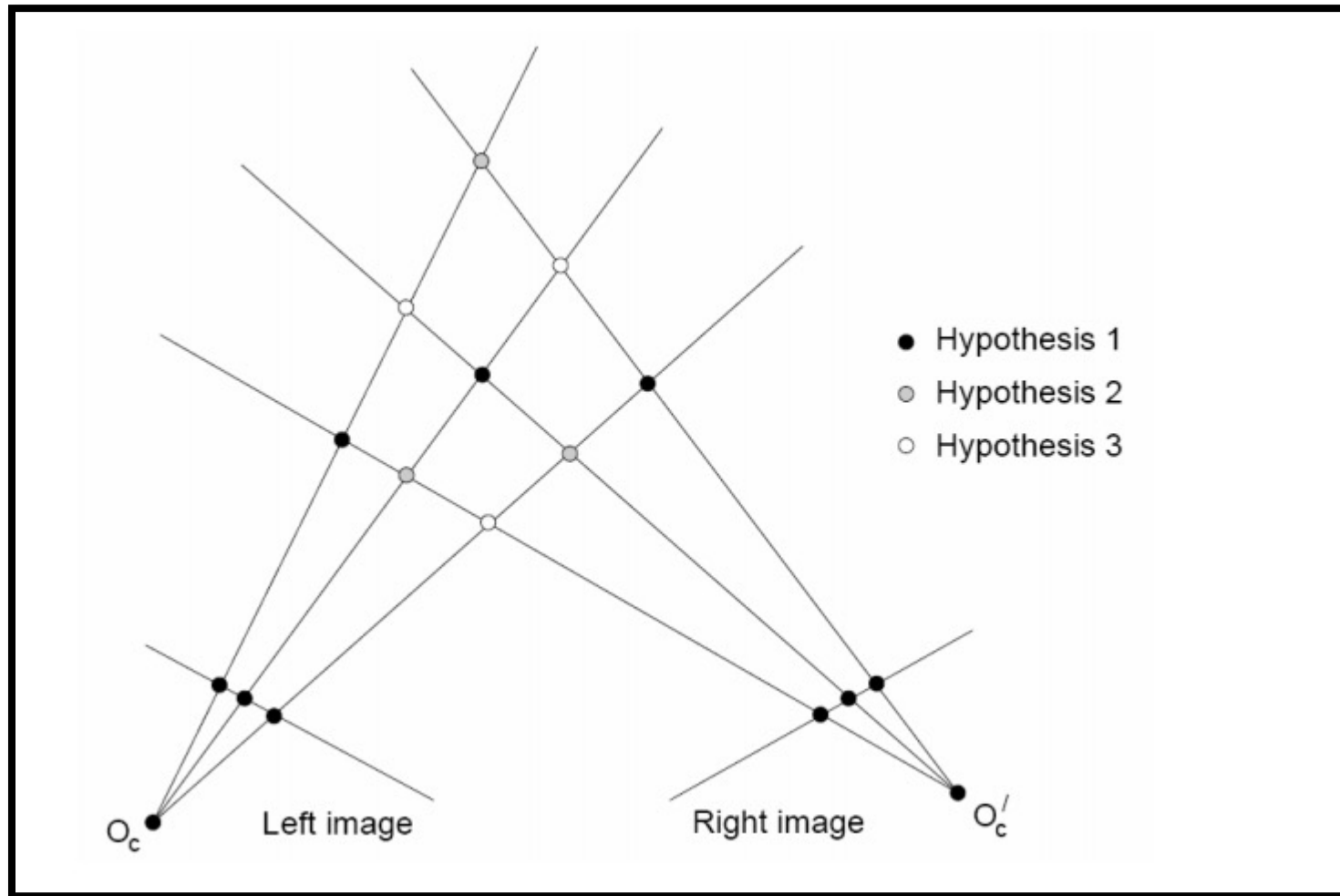
Given a calibrated binocular stereo pair, can we fuse it to produce a depth image?

Binocular Fusion Problem

- The left case, there is no ambiguity, and stereo reconstruction is a simple matter.
- In the more usual case shown on the right, any of the four points in the left picture may, match any of the four points in the right one.
- Only four of these correspondences are correct. the other ones yield the incorrect reconstructions shown as small gray discs.



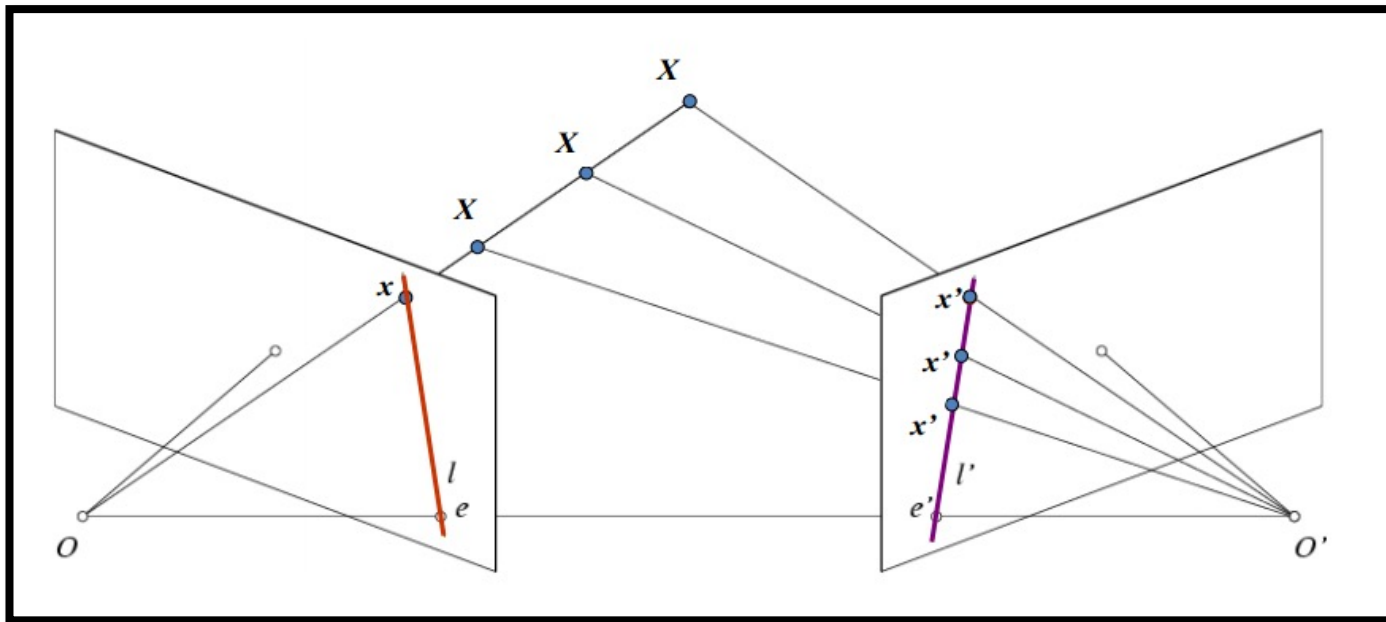
Binocular Fusion Problem



Epipolar Constraint - Point corresponds to a Ray

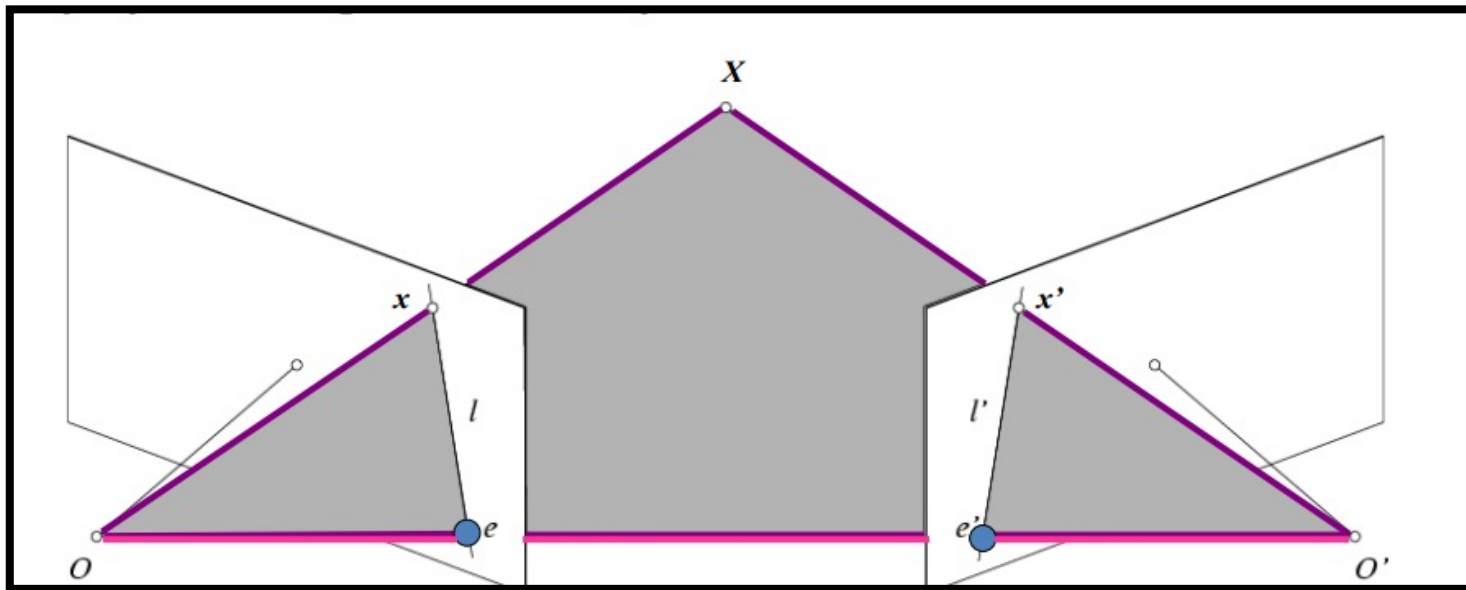
We try to estimate the position of X based on x and x' . From Geometry

- Potential matches for x have to lie on the corresponding line l' .
- Potential matches for x' have to lie on the corresponding line l .



Epipolar Geometry: Notation

- **Baseline** – Line connecting camera centers O and O'
- **Epipoles** – Intersection of baseline with the image planes, which are projections of camera centers on each other. e and e'
- **Epipolar Plane** – Planes containing baseline



Epipolar Geometry: Notation continued..

- **Epipolar Lines** - intersections of epipolar plane with image planes (always come in corresponding pairs)

In this picture l and l' are the epipolar lines, for the epipolar plane OXO'

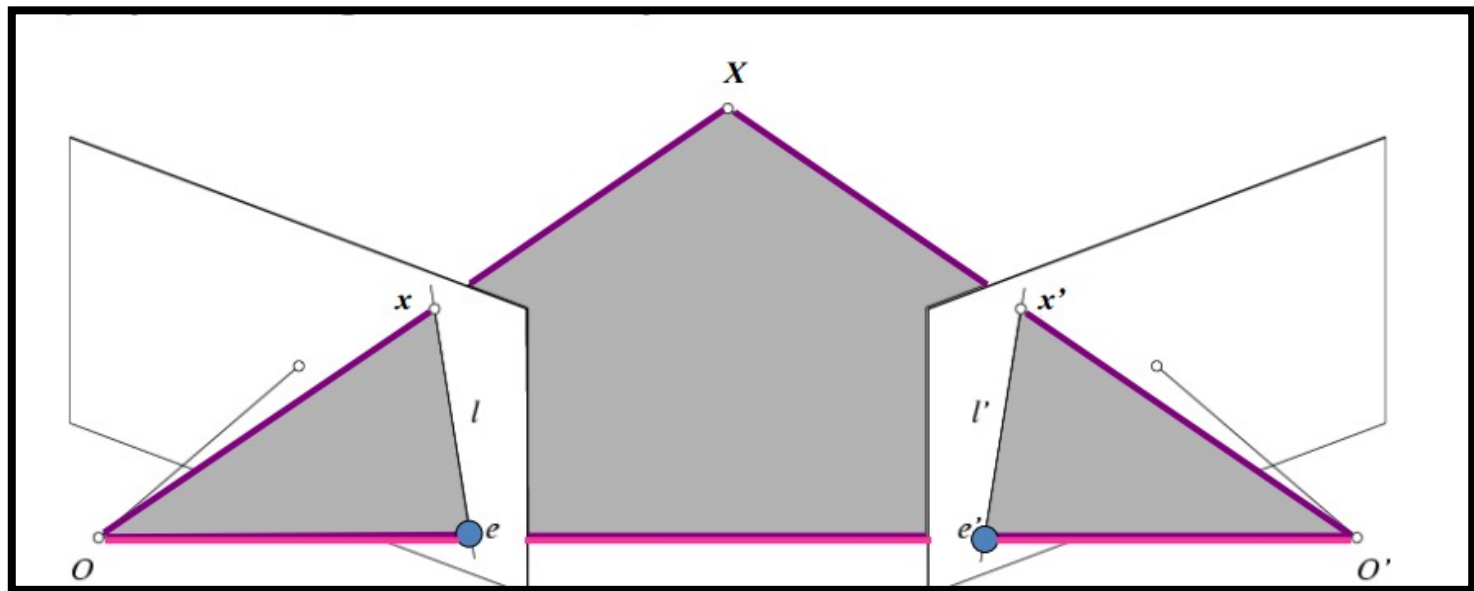


Image Rectification

- Arbitrary arrangement of cameras results in image planes that are not parallel.
- Complicated Epipolar representation and correspondence search.
How to make it simpler?

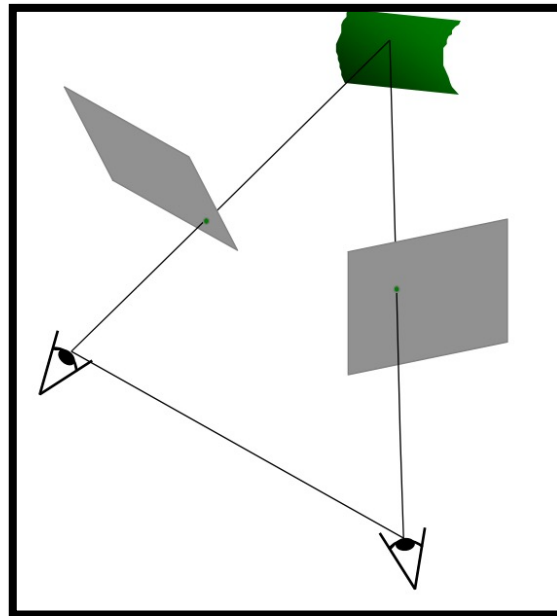
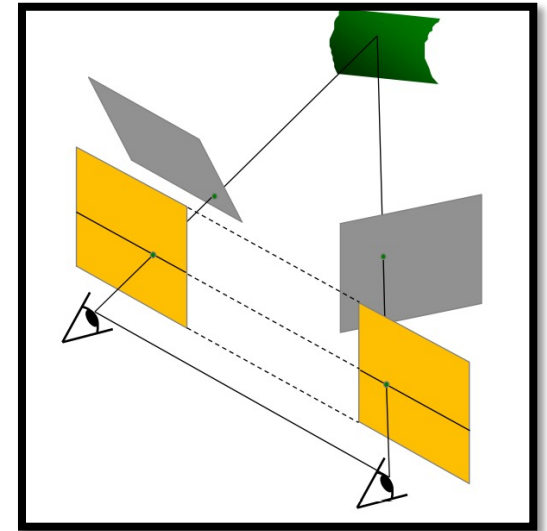


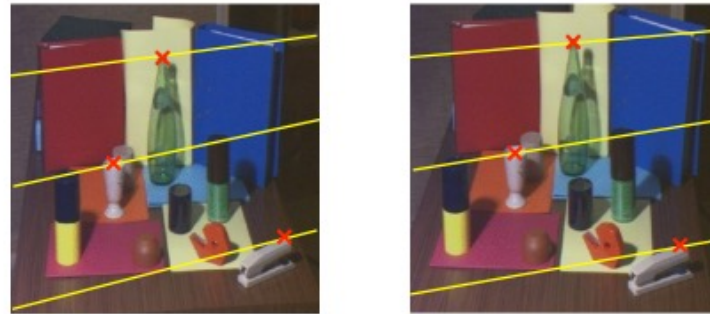
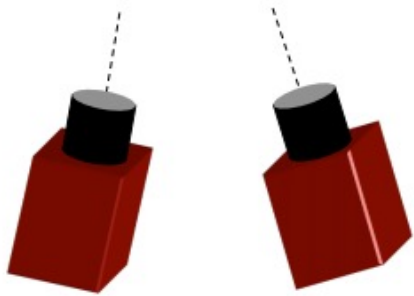
Image Rectification

- Re-project image planes onto a common plane parallel to the line between optical centers
- Pixel search would be horizontal after this transformation

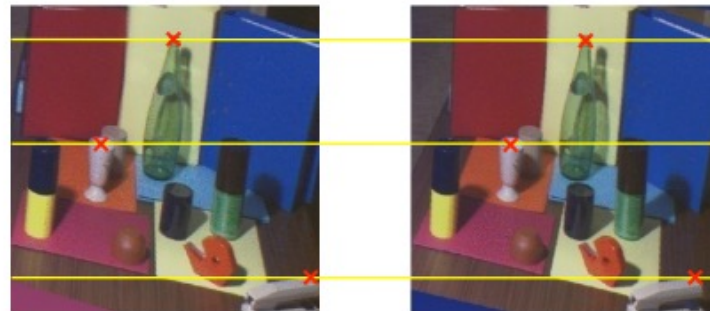
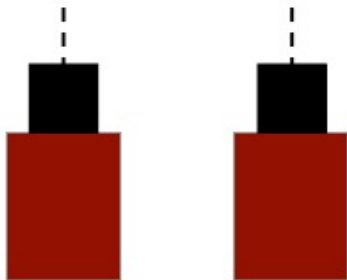


- C. Loop and Z. Zhang. Computing Rectifying Homographies for Stereo Vision. IEEE Conf. Computer Vision and Pattern Recognition, 1999

Image Rectification- Example

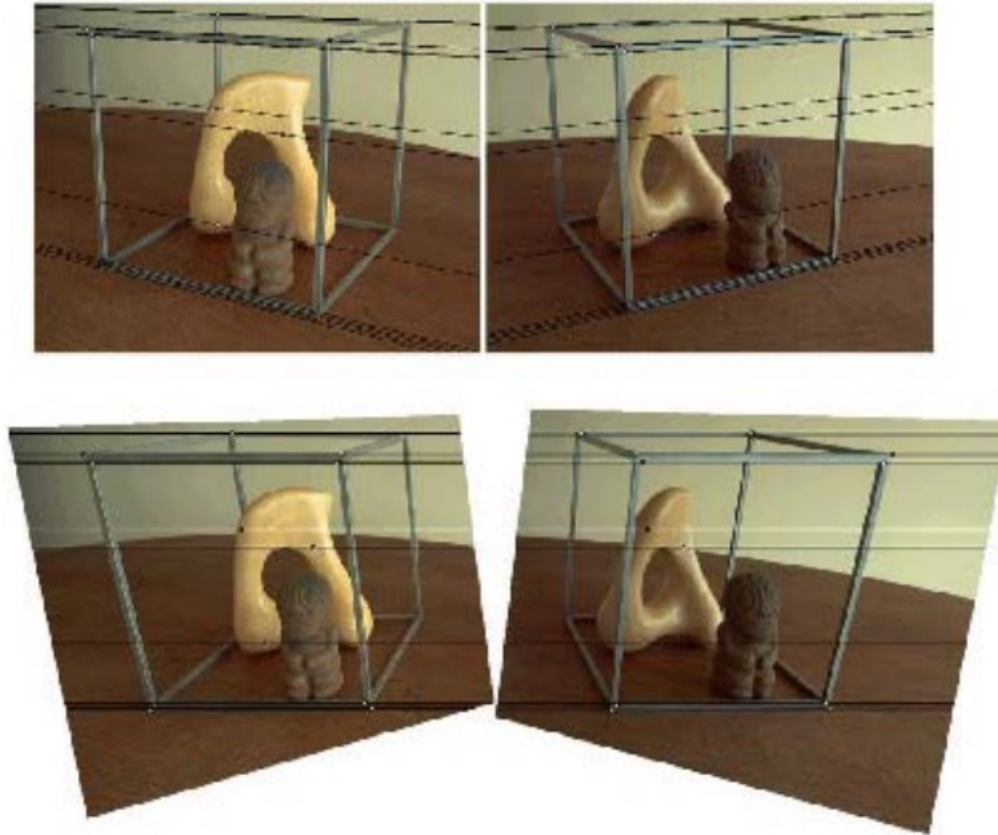


Original stereo pair

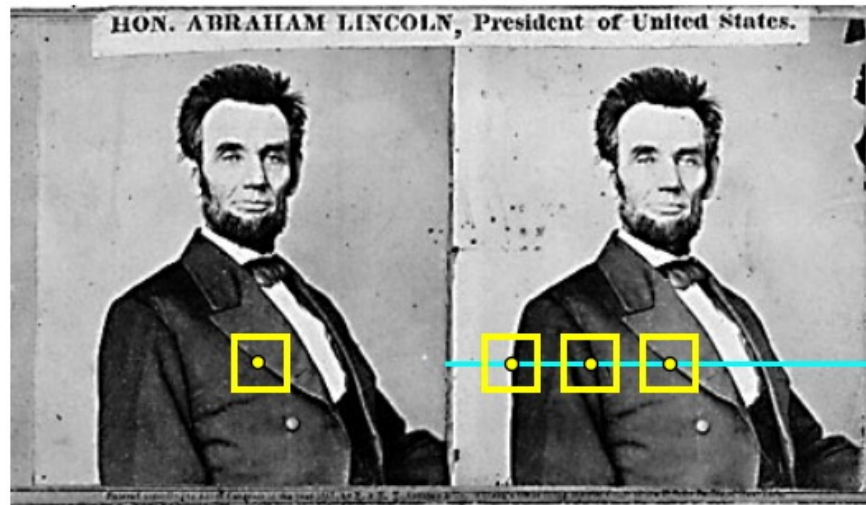


Stereo pair in standard form

Image Rectification- Example



Basic Stereo Matching Algorithm



- Perform Stereo Rectification
- For each pixel x in the first image
 - Find corresponding epipolar scanline in the right image
 - Examine all pixels on the scanline and pick the best match x'

Basic Stereo Matching Algorithm

```
for x=1:w,  
  for y=1:h,  
    bestdist=inf;  
    for i=-dr:0,  
      if (dist(pix(x,y),pix(x+i,y))<bestdist)  
        d(x,y)=i; best=sim(pix(x,y),pix(x+i,y)); end  
      end  
    end  
  end  
end
```



Advances in Computational Stereo

Myron Z. Brown, *Member, IEEE*, Darius Burschka, *Member, IEEE*, and
Gregory D. Hager, *Senior Member, IEEE*

Abstract—Extraction of three-dimensional structure of a scene from stereo images is a problem that has been studied by the computer vision community for decades. Early work focused on the fundamentals of image correspondence and stereo geometry. Stereo research has matured significantly throughout the years and many advances in computational stereo continue to be made, allowing stereo to be applied to new and more demanding problems. In this paper, we review recent advances in computational stereo, focusing primarily on three important topics: correspondence methods, methods for occlusion, and real-time implementations. Throughout, we present tables that summarize and draw distinctions among key ideas and approaches. Where available, we provide comparative analyses and we make suggestions for analyses yet to be done.

Index Terms—Computational stereo, stereo correspondence, occlusion, real-time stereo, review.

MATCH METRIC	DEFINITION
Normalized Cross-Correlation (NCC)	$\frac{\sum_{u,v} (I_1(u,v) - \bar{I}_1) \cdot (I_2(u+d,v) - \bar{I}_2)}{\sqrt{\sum_{u,v} (I_1(u,v) - \bar{I}_1)^2 \cdot (I_2(u+d,v) - \bar{I}_2)^2}}$
Sum of Squared Differences (SSD)	$\sum_{u,v} (I_1(u,v) - I_2(u+d,v))^2$
Normalized SSD	$\sum_{u,v} \left(\frac{(I_1(u,v) - \bar{I}_1)}{\sqrt{\sum_{u,v} (I_1(u,v) - \bar{I}_1)^2}} - \frac{(I_2(u+d,v) - \bar{I}_2)}{\sqrt{\sum_{u,v} (I_2(u+d,v) - \bar{I}_2)^2}} \right)^2$
Sum of Absolute Differences (SAD)	$\sum_{u,v} I_1(u,v) - I_2(u+d,v) $
Rank	$\sum_{u,v} (I_1'(u,v) - I_2'(u+d,v))$ $I_k'(u,v) = \sum_{m,n} I_k(m,n) < I_k(u,v)$
Census	$\sum_{u,v} \text{HAMMING}(I_1'(u,v), I_2'(u+d,v))$ $I_k'(u,v) = \text{BITSTRING}_{m,n}(I_k(m,n) < I_k(u,v))$

Basic Stereo Matching Algorithm

```
for x=1:w,  
  for y=1:h,  
    bestdist=inf;  
    for i=-dr:0,  
      if (dist(pix(x,y),pix(x+i,y))<bestdist)  
        d(x,y)=i; best=sim(pix(x,y),pix(x+i,y)); end  
      end  
    end  
  end  
end
```



Assumptions in Stereo Correspondence

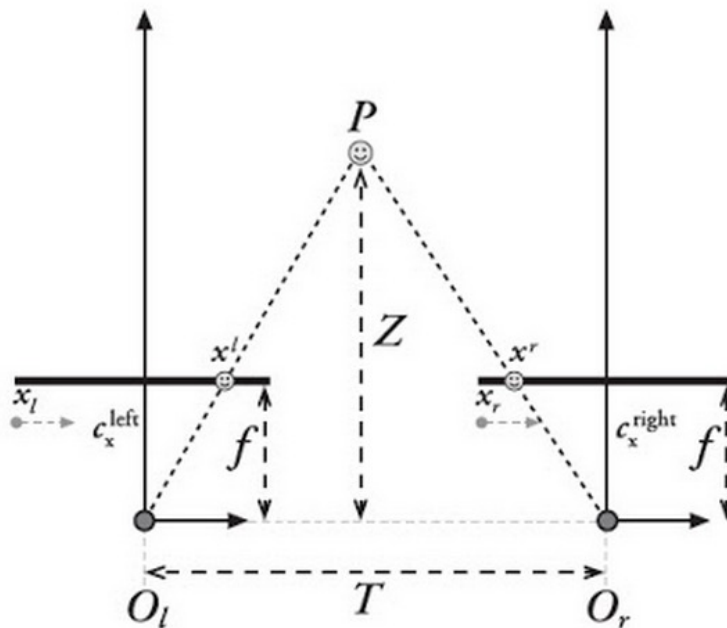
Let's make some assumptions to simplify the matching problem

1. Baseline is relatively small(compared to the depth of scene points)
2. Most scene points are visible in both views
3. Also, matching regions are similar in appearance

Disparity

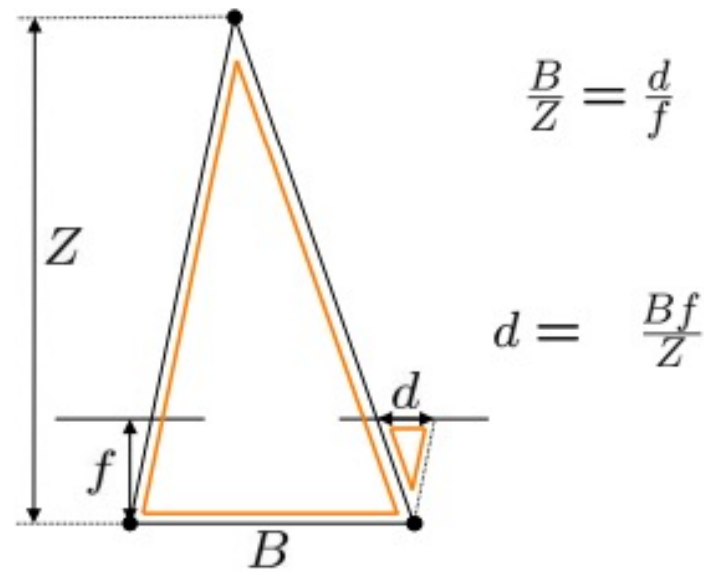
Disparity is defined as the difference in the location of a feature point between the right and left images

Disparity high for closer objects and low for distant objects



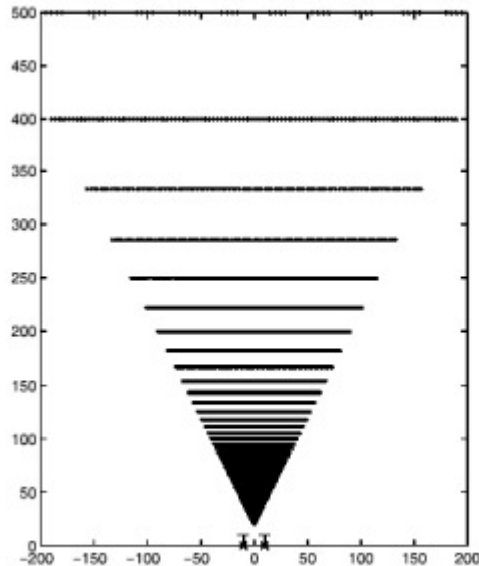
$$d = x^l - x^r$$

Depth from Disparity

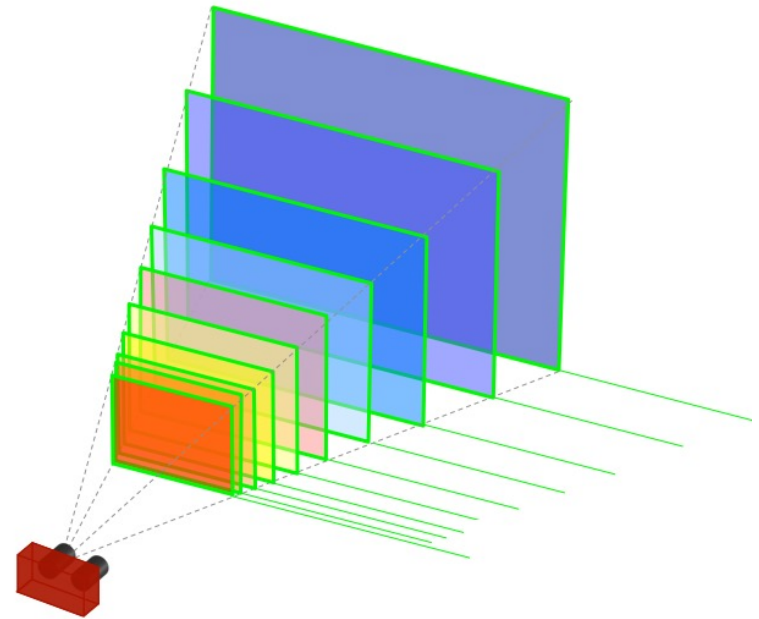


B – Baseline
Z – Depth
d – Disparity
f – focal length

Depth Resolution from Disparity



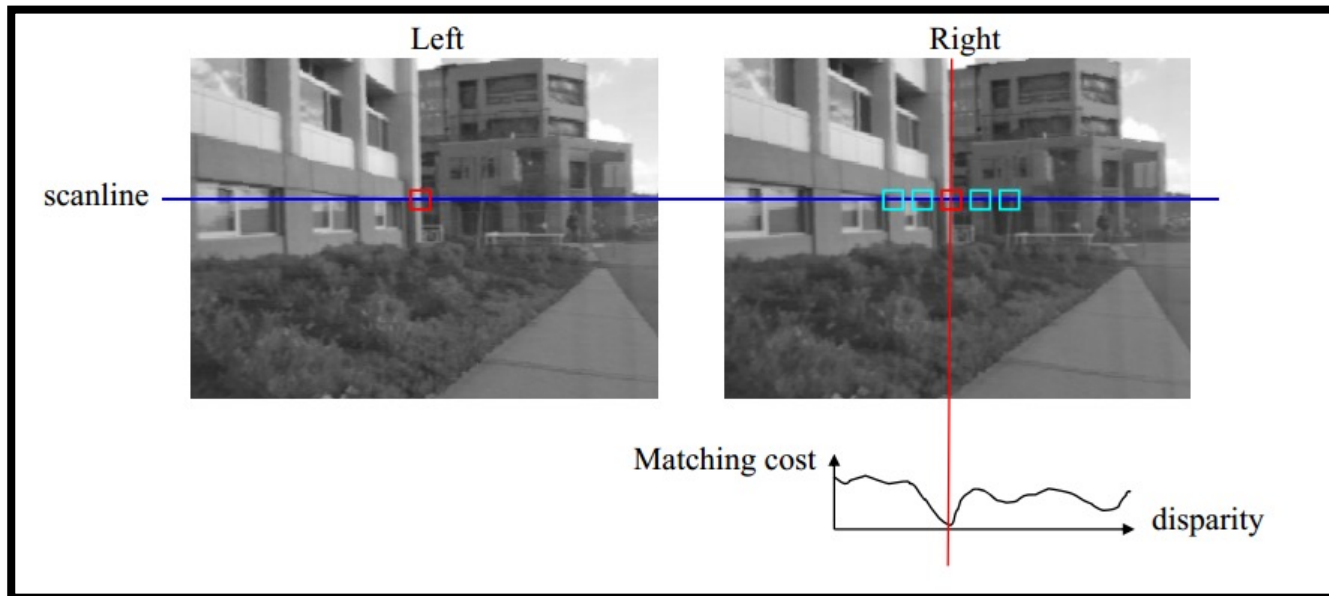
$$\frac{dd}{dZ} = \frac{Bf}{Z^2}$$



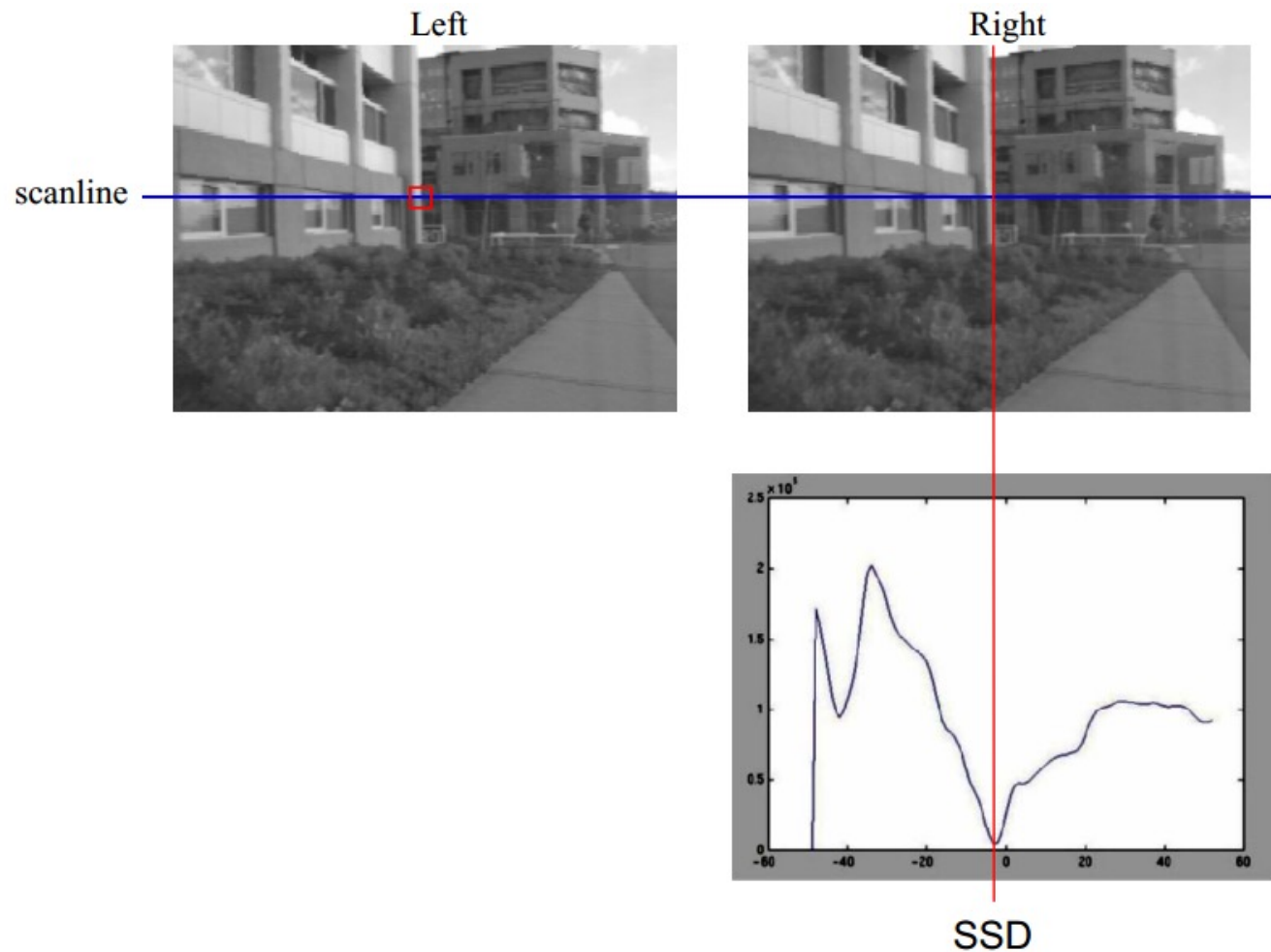
$$\Delta Z = \frac{Z^2}{Bf} \Delta d$$

Correspondence search with similarity constraint

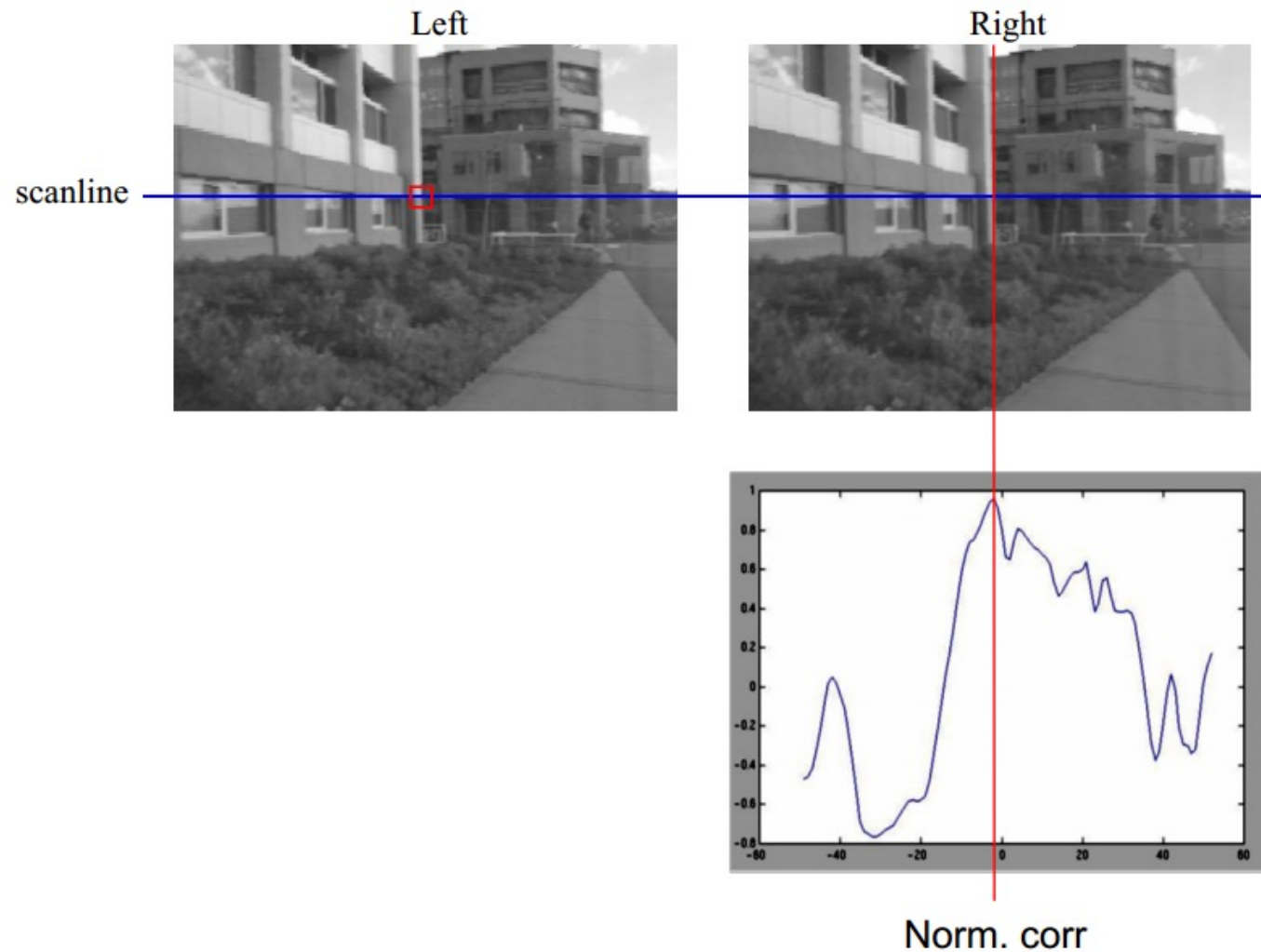
- Slide a window along the right scanline and compare contents of that window with the reference window in the left image
- Matching cost: SSD or normalized correlation



Correspondence search with similarity constraint



Correspondence search with similarity constraint

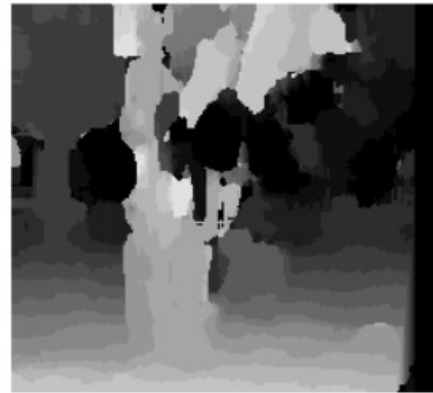


Effect of Window Size

- Smaller Window Size – More detail, But More Noise
- Large Window Size – Smooth disparity map, but less detail



$W = 3$

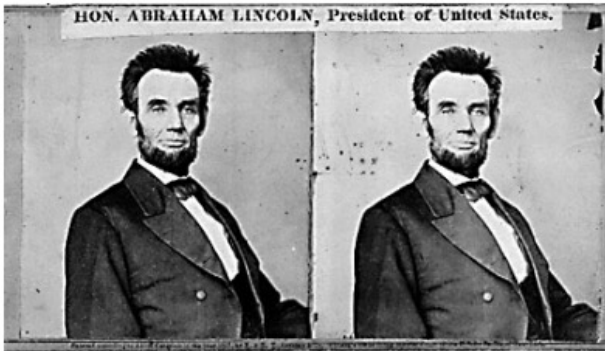


$W = 20$

Similarity Constraint

- Corresponding regions in two images should be similar in appearance
- ...and non-corresponding regions should be different
- When will the similarity constraint fail?

Similarity Constraint Failure



Textureless surfaces



Occlusions, repetition



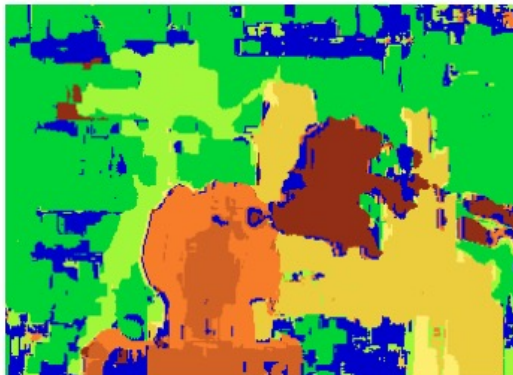
Non-Lambertian surfaces, specularities

Results with Window Search

Data



Window-based matching



Ground truth

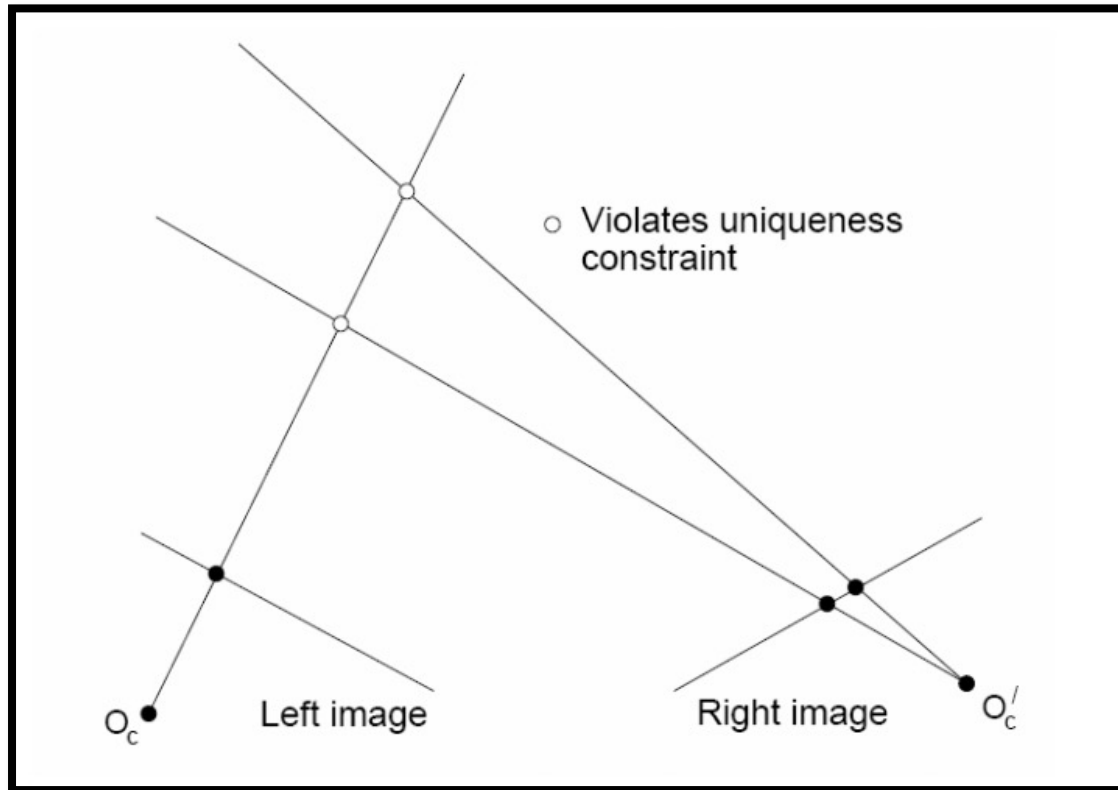


How can we improve window based matching?

- The similarity constraint is **local** (each reference window is matched independently)
- Need to enforce **non-local** correspondence constraints

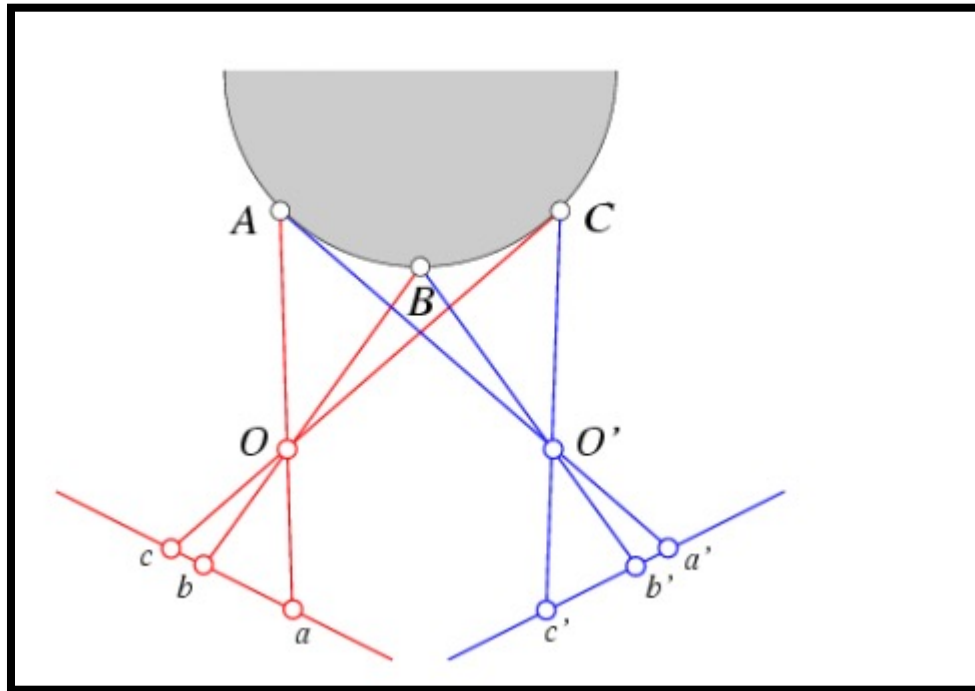
Non-Local Constraints

- Uniqueness - For any point in one image, there should be at most one matching point in the other image



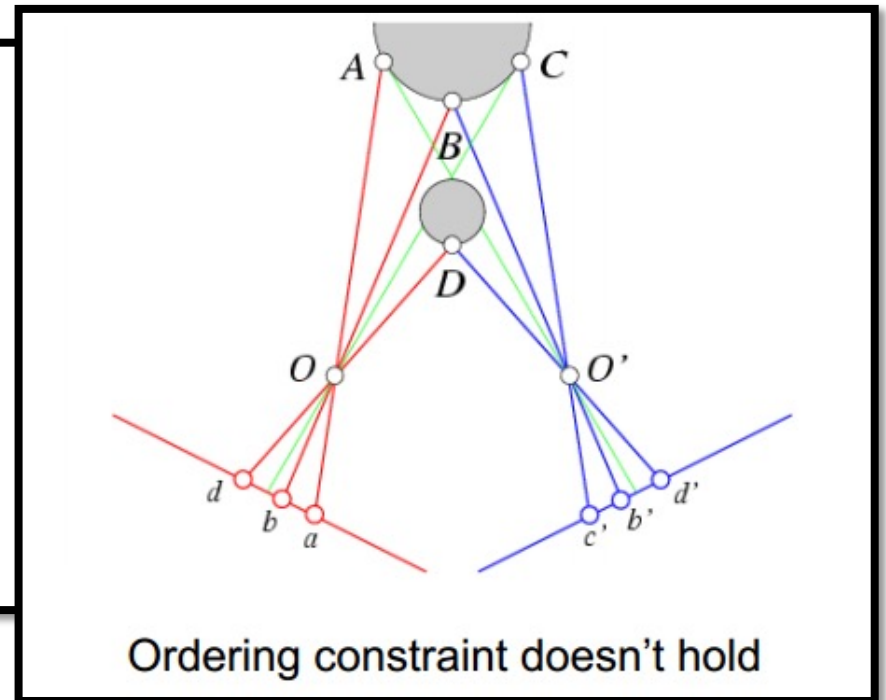
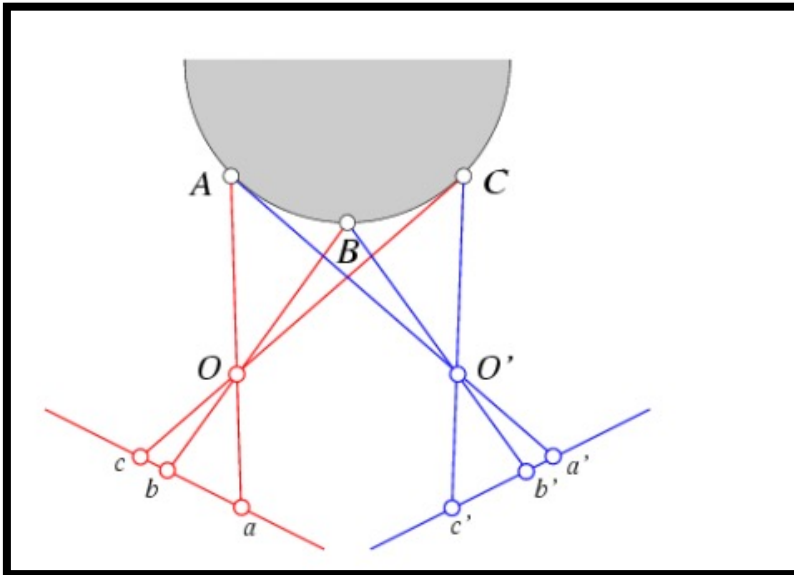
Non-Local Constraints

- Ordering - Corresponding points should be in the same order in both the images



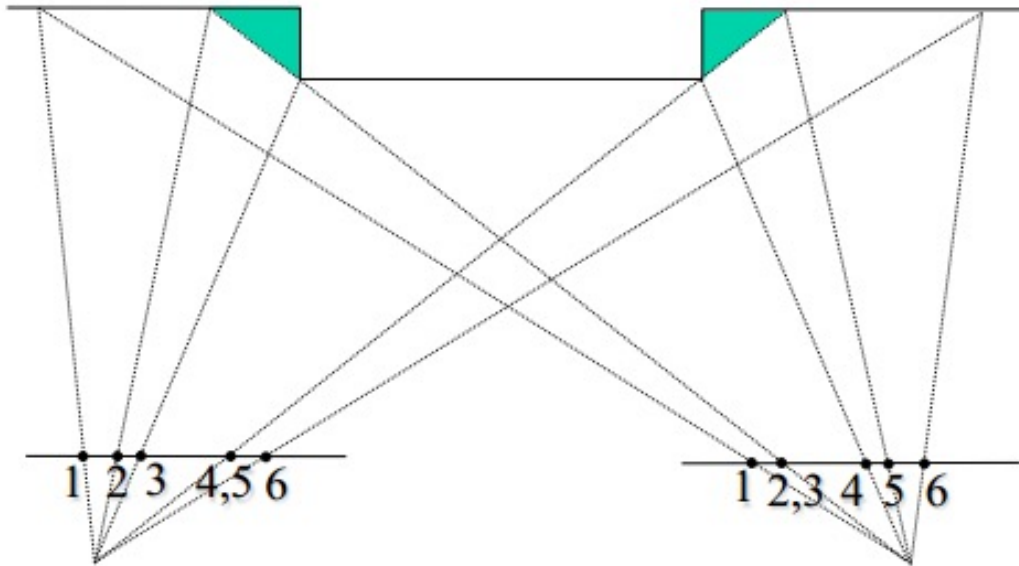
Non-Local Constraints

- Ordering - Corresponding points should be in the same order in both the images. Does not always hold true. For instance, if we have a smaller object in front of large object, the images violate ordering constraint
- Smoothness in disparity - Also We expect disparity values to change slowly

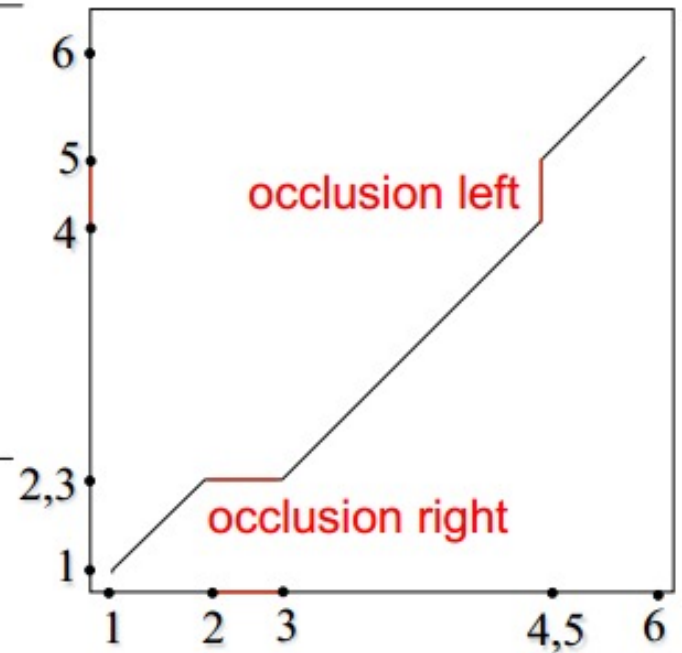


Surface as Path

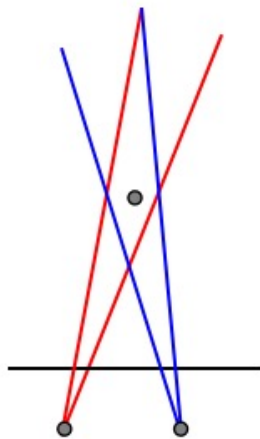
surface slice



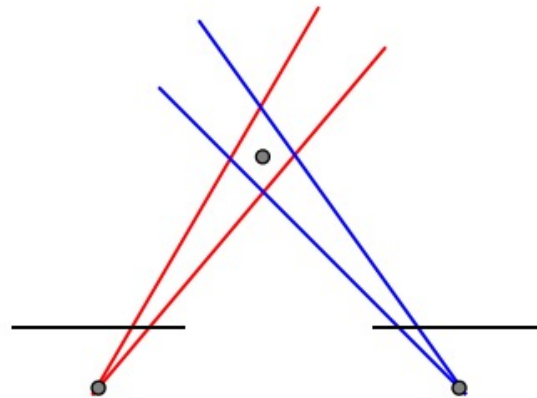
surface as a path



The role of Baseline



Small Baseline



Large Baseline

Small baseline: large depth error
Large baseline: difficult search problem

Stereo Algorithms evaluation

Middlebury Stereo Evaluation - Version 2

[New features and main differences to version 1.](#)

[Submit and evaluate your own results.](#)

☐ Open a new window for each link

Error Threshold = 1 Error Threshold... ▾		Sort by nonocc ▼			Sort by all ▼			Sort by disc ▼			Average percent of bad pixels (explanation)			
Algorithm	Avg.	<u>Tsukuba</u> ground truth			<u>Venus</u> ground truth			<u>Teddy</u> ground truth				<u>Cones</u> ground truth		
	Rank ▼	nonocc	all ▼	disc	nonocc	all ▼	disc	nonocc	all ▼	disc		nonocc	all ▼	disc
TSGO [143]	10.5	0.87 4	1.13 1	4.66 8	0.11 7	0.24 9	1.47 10	5.61 38	8.09 17	13.8 31	1.67 1	6.16 1	4.95 1	4.06
JSOSP+GCP [151]	11.8	0.74 1	1.34 8	3.98 1	0.08 2	0.16 1	1.15 2	3.96 14	10.1 33	11.8 17	2.28 15	7.91 29	6.74 18	4.18
ADCensus [82]	14.1	1.07 19	1.48 17	5.73 22	0.09 3	0.25 12	1.15 2	4.10 16	6.22 7	10.9 13	2.42 21	7.25 15	6.95 22	3.97
AdaptingBP [16]	17.9	1.11 22	1.37 10	5.79 24	0.10 5	0.21 8	1.44 9	4.22 18	7.06 15	11.8 18	2.48 25	7.92 31	7.32 30	4.23
CoopRegion [39]	18.4	0.87 6	1.16 2	4.61 5	0.11 6	0.21 6	1.54 14	5.16 30	8.31 20	13.0 25	2.79 43	7.18 14	8.01 50	4.41
CCRADAR [152]	22.5	1.15 25	1.42 15	6.23 38	0.15 19	0.27 15	1.89 24	5.39 33	10.6 38	14.7 42	2.01 2	7.37 17	5.88 2	4.75
RDP [87]	23.8	0.97 11	1.39 12	5.00 11	0.21 40	0.38 31	1.89 24	4.84 22	9.94 32	12.6 22	2.53 29	7.69 21	7.38 31	4.57
MultiRBF [129]	24.2	1.33 49	1.56 22	6.02 33	0.13 11	0.17 3	1.84 21	5.09 28	6.36 8	13.4 29	2.90 51	6.76 8	7.10 27	4.39
DoubleBP [34]	24.9	0.88 8	1.29 6	4.76 9	0.13 12	0.45 49	1.87 23	3.53 12	8.30 19	9.63 8	2.90 50	8.78 61	7.79 42	4.19

<http://vision.middlebury.edu/stereo/eval/>